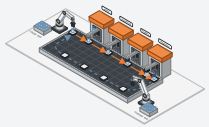
Asset | AAS: the desired state | Tools | Running system

Line



FillingLine
Stoppering | Capping
Filling | Inspection

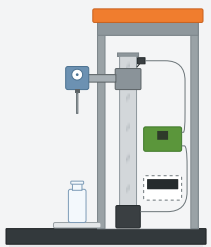
Vial2mL ❶
requires **Filling**:
FillVolume 2.0 mL,
result FillWeight

❷ Match capabilities
offered
offered after a reconfiguration
not offered

offered

❺ Bind and run
The orchestrator (agents or a behaviour tree) occupies the module and calls Start(Session, Volume)

Module



FillingModule
Capability Description | Interfaces
Skills: **Dispensing**, Built | PlannedFilling, *Described*
NeedleDown → Dispense → Home → Weigh

Described

❸ Author a skill
written in the skill editor,
or **planned** from capabilities

desc(A)
Built

❹ Reconcile
Compare read(Π) with desc(A)
Check primitives, limits
Guard values only when idle
Apply management commands
Verify read(Π) = desc(A)
Record boot file; Built

apply
read(Π)

FORTE, Pi 5 | OPC UA
Occupation | PackML
Axis | Pump | Scale
Dispensing *running*
PlannedFilling *new*
composites: instances only

Components



MoveAxis | Home
LinearAxis

Dispense
Pump

Tare | Weigh
Scale

Baseline
a new boot file and a restart:
the module stops

Primitive block types
SK_MoveAxis, SK_Dispense, …
built into FORTE with drivers

change: IEC 61499 management
operate: OPC UA
a step runs a primitive
part of; the baseline